Byte-ing Off More Than You Can Chew

Reimplementing **Byte Latent Transformer** (Pagnoni et al., 2024). **ACL 2024 Outstanding Paper.**

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Problems with Traditional Tokenization

Static Compute

Author of Harry Potter is → ?

Patient's diagnosis is → ?

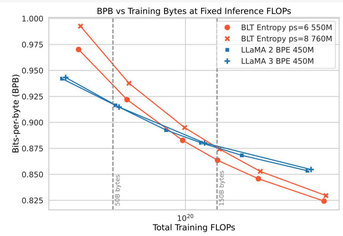
Character Awareness

add a character 'z'

not → znotz

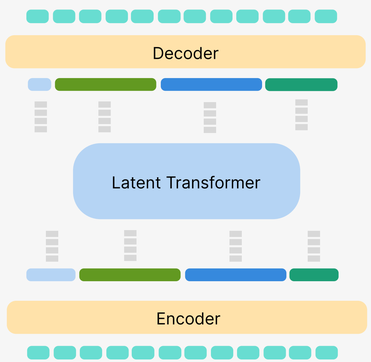
Solution: Entropy Patching

Goal of the Project

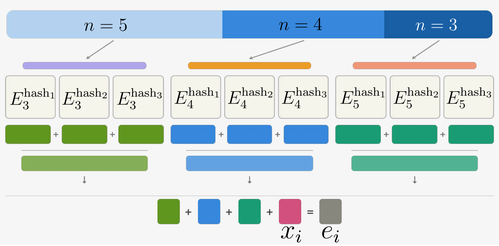


Goal: At a much smaller scale, show that BLT can overtake BPE-tokenizer-based LLM BPB as shown in the paper

Methodology



$$e_i = x_i + \sum_{n=3,\ldots,8} \sum_{j=1,2,3} E_n^{\text{hash}_j}(\text{Hash}_j(g_{i,n}))$$



Design Choices

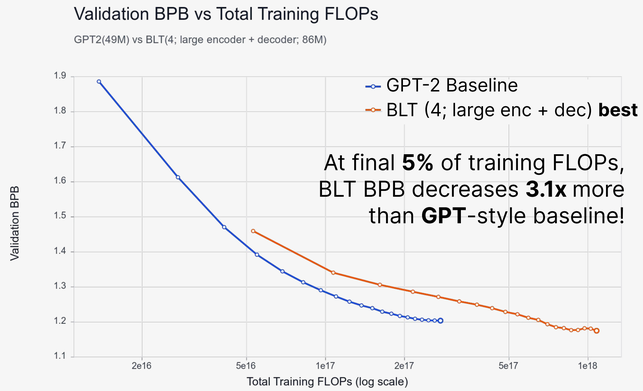
Fewer, bigger patches at inference lets us train a larger model under matched FLOPs.

DATA	COMPUTE
FineWeb HF	375 H100 GPU-hours
1.7B TRAIN 3.4M VALIDATION	served via Prime Intellect

Experiments

BPB	Variant	N-gram	Params
1.22	GPT-2 Baseline	BPE	49M
1.27	Small Encoder **(Baseline)**	3, 4, 5	86M
1.31	Small Decoder **(Adversarial)**	4	86M
1.21	Large Encoder + Decoder	3, 4, 5	86M
1.21	Large Encoder + Decoder **(Best)**	4	86M
1.26	+ Patch Length Embeddings	4	86M

Results

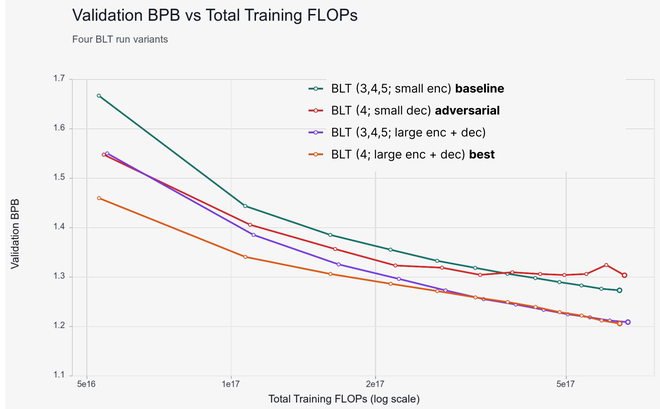


Using the same data, we achieve a lower BPB!

References

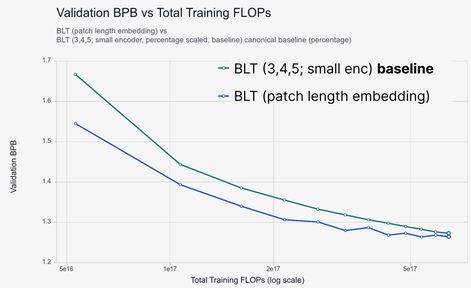
Pagnoni, A., Pasunuru, R., Rodriguez, P., Nguyen, J., Muller, B., Li, M., Zhou, C., Yu, L., Weston, J., Zettlemoyer, L., Ghosh, G., Lewis, M., Holtzman, A., & Iyer, S. (2025). Byte Latent Transformer: Patches Scale Better Than Tokens. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 9238–9258.

Q: At an ultralow parameter scale, what heuristics change?



Result 1: Optimal parameter allocation differs!
Result 2: Not all n-gram context is made equal.

What if we add patch length embeddings?



Conclusion

- Reproducing byte-level transformers at ultralow parameter counts requires substantial **parameter re-allocation** for complex architectures like BLT.
- We introduce context embedding simplifications and a new patch length embedding which improves the paper's canonical architecture.